Name:		Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Exam
Year 6

H2 Biology

9744/04

Paper 4 Practical

24 Aug 2017
2 hour 30 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, class index number and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.

Answer **all** questions.

Write your answers in the space provided in the question paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

Give details of the practical shift and laboratory in the boxes provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Shift	Laboratory

FOR EXAMINER'S USE Score	
1	/20
2	/20
3	/15
Total	/55

This document consists of **14** printed pages (including this cover page) and **0** blank page.

List of Apparatus and Materials

Please note that items **1** – **5** are only available to you for the first 75 minutes of this examination.

PLACE ITEM **1-5** BACK IN THE BASKET AFTER 75 MINUTES.

Item	**Apparatus / Reagents / Chemicals**	**Quantity**	**Time allocation**
1	Toothpick	2	0 – 75 minutes
2	Clean slide	2	
3	Cover slip	2	
4	Sun shade slide	1	
5	Stage micrometer 0.01mm	1	
6	Pipe cleaner	4	
7	White wire	1	
8	Label	4	
9	Transparent tape	1	

Item	**Apparatus / Reagents / Chemicals**	**Quantity**
10	Dried yeast	1 vial
11	Glucose powder	1 vial
12	250cm^3 beaker	1
13	Boiling tube	2
14	Bung and delivery tube	2
15	Test tube	2
16	Test-tube rack	1
17	10 cm^3 syringe	1
18	Styrofoam cup	1
19	Glass rod	1
20	Thermometer	1
21	Stop watch	1
22	Marker pen	1

Question 1

For Examiner's use

Prepare wet mount of your cheek cells.

Carefully use a toothpick to scrape the inside of your cheek. Transfer the scraping directly onto a clean slide. Add a very small drop of water onto the slide. Then cover with the cover slip.

(a) (i) Make a large, labelled, high-power drawing to show three cheek cells. [3]

(ii) Describe the procedure, including the use of the eyepiece graticule, to determine the ratio between the diameter of cheek cells and that of their nuclei from (a)(i). [4]

You are given a microscope slide of a sun leaf. Examine the slide under the high-power objective lens of your microscope and locate palisade mesophyll cells.

(b) (i) Make a detail, labelled drawing in the space below of three adjacent cells. [3]

(ii) Using the eyepiece graticule fitted in the eyepiece lens of your microscope, and the stage micrometer, find the actual length, in μm, of one of the cells that you have drawn.

Show the measurements that you have made and your working. [4]

Length of one of the cells = ___________________ μm

For Examiner's use

(c) Use the pipecleaners with which you have been provided to construct a model which shows the structure of a bivalent (i.e. a pair of homologous chromosomes at the end of the prophase stage of meiosis I).

(i) Use appropriately coloured pipecleaners to represent the parts of the bivalent derived from each different chromosome of the homologous pair and white wire to represent centromeres. [2]

(ii) Show one chiasma on your model by twisting the pipecleaners around each other. [1]

Assume that two different genes, A and B, are carried on these chromosomes and that the organism is heterozygous at these loci.

(iii) Write down the genotype of this organism. [1]

____

(iv) Add appropriate sticky labels to your model to show a possible location for the relevant alleles if the gametes which result from this meiosis are of four different genotypes. [2]

At the end of the examination, use the transparent tape provided to securely stick your model in the space below.

[Total: 20 marks]

For Examiner's use

Question 2

Enzymes in respiring yeast convert glucose to carbon dioxide and water. You are required to investigate the effects of temperature on these enzymes by measuring the rate of carbon dioxide production.

Prepare a water-bath by half-filling a 250 cm^3 beaker with water and **maintain is temperature** between 35 ºC and 40 ºC.

Label two boiling tubes **T1** and **T2** respectively.

You are provided with 1 vial of dried yeast, and 1 vial of glucose powder. Add them into boiling tube **T1** and add 10 cm^3 of water to make the yeast suspension. Stir thoroughly with a glass rod until you get a homogenous yeast suspension. Fit the bung and delivery tube provided.

Add 10 cm^3 of tap water to **T2** and fit the bung and delivery tube provided. Ensure that the bungs are of a sufficiently tight fit to make them airtight.

Place the boiling tubes in the water-bath with the delivery tubes in test-tubes of cold water, as shown in **Fig. 1.1**.

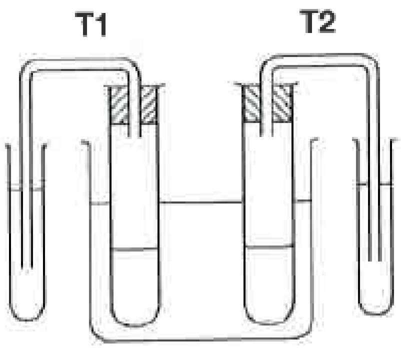

Fig. 1.1

You will soon notice bubbles of gas appearing from the ends of the delivery tubes.

Wait three minutes.

For Examiner's use

(a) (i) Count the number of bubbles produced by **T1** in 30 seconds. Enter your reading in **Table 1.1**. Wait 30 seconds and then count the bubbles produced by **T2** in the next 30 seconds. Repeat this procedure twice more. This will give you three readings for each boiling tube.

For Examiner's use

Table 1.1

	bubbles / 30 sec			
	35 ºC – 40 ºC		45 ºC – 50 ºC	
reading	**T1**	**T2**	**T1**	**T2**
1				
2				
3				
mean				

(ii) Increase the temperature of the water-bath to between 45 ºC and 50 ºC and repeat the experiment.

Enter your readings in **Table 1.1**.

(iii) Calculate the mean bubbling rates for all four sets of figures and enter your results in **Table 1.1**. [4]

(b) Explain why you waited three minutes before counting the gas bubbles. [2]

(c) With reference to your results, explain the effect of raising the temperature of **T1** to between 45 ºC and 50 ºC. [4]

(d) State the purpose of **T2** in your experiment and explain how it could be used to make your results more reliable. [2]

(e) Explain how you could improve the procedure even further to ensure that your results were even more reliable. [3]

For Examiner's use

For Examiner's use

(f) In another experiment, different carbohydrate sources were added to the yeast suspension. The amount of gas produced in 20 minutes was measured at 35 ºC – 40 ºC. The results are shown in table 1.2.

Table 1.2

Carbohydrate added	Amount of gas emitted in 20 min / cm^3	
glucose	2.7	
sucrose	2.9	
lactose	0.0	

(i) Calculate the rate of gas production and enter your results into Table 1.2. [2]

(ii) Explain the results. [3]

[Total: 20 marks]

Question 3

For Examiner's use

The enzyme amylase breaks down starch into maltose. There are several types of amylases, such as α, β and γ amylase which are used in the food and beverage industry.

Two newly identified α amylase and β amylase have yet to be tested for their activities at a range of enzyme concentration and temperatures. The optimal enzyme concentration is the lowest concentration of enzyme needed to reach the maximum rate of reaction.

Using your own knowledge, design an experiment to determine the greatest rate of activity for each of the two amylases α and β when used at its optimal enzyme concentration and temperature when subjected to the same pH.

Your planning must be based on the assumption that you have been provided with the following equipment and materials, which you must use:

- 5% α amylase,
- 5% β amylase,

- Distilled water,
- 2% starch suspension,
- 3% Iodine solution,
- timer, e.g. stopwatch
- thermostatically controlled water-bath and thermometer.

You may select from the following apparatus and use appropriate additional apparatus:

- a variety of different sized beakers, measuring cylinders or syringes for measuring volumes.
- a range of buffer solutions between pH 2 and pH 9,
- pH probe and digital meter.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 15 marks]

For Examiner's use

END OF PAPER

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Paper 4 Practical

Answer Scheme

1
(a)(i)

- 3 large cheek cells with nucleus drawn
- drawing quality – clear continuous line with no shading, cell wall show as double line
- Label – cytoplasm, nucleus, membrane (at least 3)

(a)(ii)

- Record number of eyepiece unit of the 3 cheek cells from (i)
- Record number of eyepiece unit of their nuclei
- Find average of both
- Find ratio

(b)(i)

- 3 adjacent palisade mesophyll cells drawn
- drawing quality – clear continuous line with no shading, cell wall show as double line
- Label – cytoplasm, nucleus, membrane, chloroplasts, vacuole (at least 3)

(b)(i)
Calibration
Length of cell in eyepiece unit
Actual length
Answer in μm

(c)
(i)

- Homologous chromosomes represented by 2 sets of same colour wires
- Centrosome at same position

(ii)
Chiasma

(iii)
AaBb

(iv)
- AA/aa/BB/bb on same chromosome (same X)
- B/b below crossing over

2 (a) (iii)

Values for T2 columns stated as 0
Values for T1 (35-40 ºC) column lower than T1 (45-50 ºC)
Mean values correctly calculated to whole number

1 (b)

To allow content in the boiling tubes to acclimatize to the temperature in the water bath
This is to ensure the that number of bubbles counted is representative of the rate of respiration at that temperature

1 (c)
Quote data
Hence with raising the temperature from 35-40 ºC to 45-50 ºC, the rate of respiration increased
More CO_2 is released

increased temperature, the enzymes have higher kinetic energy. Rate of effective collision between the enzymes and its substrates are higher and hence there are more enzymes-substrate complexes

1 (d)
T2 is a negative control to show that the set-up itself will not emit gas / cause bubbles to be formed
Instead of tap water, 20 cm^3 of water with the same amount of glucose added to yeast suspension S1 should be used. This ensures that presence of yeast is the only dependent variable between T1 and T2.

1 (e)
At step (a) (ii), new yeast suspension should be used
After performing the experiment on the yeast suspension at 35-40 ºC, some glucose in the suspension would be used for respiration, changing the concentration of glucose
As a result, glucose concentration might become limiting when rate of respiration is measured on the same yeast suspension at 45-50 ºC, causing the measured rate to be lower than the actual rate.

1 (f) (i)
Appropriate heading with units (e.g. rate of gas production / $cm^3 min^{-1}$) ;
Correctly calculated rate recorded to 3sf

1 (f) (ii)
glucose is the primary respiratory substrate
Yeast is able to respire with sucrose because it has the enzyme sucrose to hydrolyse the disaccharide into glucose and fructose, which are monosaccharides
Yeast is unable to respire lactose because it has no enzyme lactase to hydrolyse the disaccharide to monosaccharides glucose and galactose

3. Planning

Theory	Enzymes such as amylase are often made of proteins. Enzyme concentration determines the amount of active site available for formation of enzyme-substrate complex and hence the rate of reaction. Temperature affects the structure and shape of active site of the enzyme and hence the rate of reaction. At low concentrations of enzyme, enzyme is the limiting factor as there are more active sites than substrate. As enzyme concentration increases, the number of active sites available for formation of enzyme-substrate complex increases and rate of reaction increases. At high concentrations of enzyme, substrate is the limiting factor as there are empty active sites. As temperature increases towards the optimum temperature, the rate of reaction would increase, doubling for every increase of 10^0C. The rate of reaction of an enzyme is highest at its optimum temperature. After the optimum temperature, the rate would decrease sharply as the hydrogen and ionic bonds in the enzyme's active site are broken by the increase in kinetic energy and the enzyme is said to be denatured. The rate of reaction within each factor (enzyme concentration and temperature) can be compared by the amount of time required to digest starch into maltose which is indicated by the colour change of iodine from blue-black to yellow. The shortest time indicate fastest rate. After obtaining the optimum of each factor for both enzymes, the experiment is carried out at each enzyme's optimum enzyme concentration and temperature to obtain its highest rate of reaction.
Variables	Independent variable to obtain optimum enzyme concentration: 5 enzyme concentrations of : 1, 2, 3, 4, 5% Independent variable to obtain optimum temperature: 5 temperatures of: 10, 20, 30, 40, 50^0C. Dependent variables: Time taken for iodine to change from blue-black to yellow. Controlled variables for all experiments: pH of reaction mixture, kept constant at pH 7 Volume of substrate starch solution, kept constant at 5 cm^3 Volume of α and β amylase used in reaction, keep constant at 1 cm^3
Controls	A negative control can be set up by replacing α and β amylase with distilled water and carrying out the experiment.
Procedure	To determine optimum enzyme concentration: 1. Prepare 5 test tubes with 5 cm^3 of 2% starch suspension each using a 5 cm^3 syringe.

2. Prepare enzyme solutions of 5 different concentrations using simple dilution according to the table below. Label the beakers accordingly.

Label	Concentration of α amylase / %	Volume of 5% α amylase stock / cm^3	Volume of distilled water / cm^3	Total volume / cm^3
α1	5	10.0	0.0	10.0
α2	4	8.0	2.0	10.0
α3	3	6.0	4.0	10.0
α4	2	4.0	6.0	10.0
α5	1	2.0	8.0	10.0

3. Add 2 cm^3 of pH 7 buffer using 5 cm^3 syringe to the mixture and ensure that the pH is constant by monitoring with the pH meter.
4. Add 1 cm^3 of the enzyme into the substrate and immediately start the stop watch.
5. After every 1 minute, remove 0.5 cm^3 of mixture for iodine test.
6. Put the mixture on a white tile and add 3 drops of iodine using a dropper onto the mixture and mix well.
7. Record the time it takes for iodine test to give a yellow result in the table below.

Enzyme α amylase				
Enzyme concentration	Time taken for iodine to turn yellow / s			
	Reading 1	Reading 2	Reading 3	Average
5				
4				
3				
2				
1				

8. Repeat steps 1- 7 for enzyme β amylase.
9. Conduct 2 replicates and 3 repeats for both enzymes to ensure consistency and reproducibility.

For determining optimum temperature:

1. Prepare 5 test tubes with 5 cm^3 of 2% starch suspension each using a 5 cm^3 syringe.
2. Add 2 cm^3 of pH 7 buffer using 5 cm^3 syringe to the mixture and ensure that the pH is constant by monitoring with the pH meter.
3. Prepare 10 cm^3 of 5% enzyme using a 10 cm^3 syringe.
4. Equilibrate temperature of the substrate and enzyme by putting them into the thermostatically-controlled water bath of 10^{o}C for 5 minutes.
5. Measure the temperature of substrate and enzyme solution using the thermometer to check the desired temperature is reached.
6. Add 1 cm^3 of the enzyme into the substrate and immediately start the stop watch. Ensure that test tube with mixture is in water bath.
7. After every 1 minute, remove 0.5 cm^3 of mixture for iodine test.
8. Put the mixture on a white tile and add 3 drops of iodine using a dropper onto the mixture and mix well.
9. Record the time it takes for iodine test to give a yellow result in the table below.

Enzyme α amylase				
Temperature /°C	Time taken for iodine to turn yellow / s			
	Reading 1	Reading 2	Reading 3	Average
10				
20				
30				
40				
50				

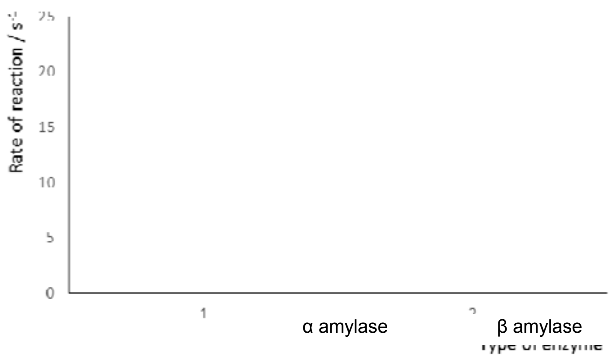
10. Repeat steps 1 – 9 for enzyme β amylase.
11. Conduct 2 replicates and 3 repeats for both enzymes to ensure consistency and reproducibility.

To determine the greatest rate of activity for each enzyme:

1. Prepare 1 test tube with $5 cm^3$ of 2% starch suspension.
2. Add $2 cm^3$ of pH 7 buffer and ensure that the pH is constant.
3. Prepare 1 test tube with $5 cm^3$ of optimum concentration of α amylase as determined earlier.
4. Equilibrate both enzyme and substrate to the optimum temperature determined earlier by using the water bath.
5. Add $1 cm^3$ of enzyme into the substrate and immediately start the stop watch.
6. After every 1 minute, remove $0.5 cm^3$ of mixture for iodine test.
7. Put the mixture on a white tile and add 3 drops of iodine using a dropper onto the mixture and mix well.
8. Record the time it takes for iodine test to give a yellow result in the table below.
9. Repeat steps 1 – 8 for enzyme β amylase.

	Time taken for iodine to turn yellow / s				
	Reading 1	Reading 2	Reading 3	Average	Rate of reaction / s^{-1}
α amylase					
β amylase					

10. Plot the bar graph of rate of reaction against each type of enzyme.



Prep List (Confidential)

Item	Apparatus / Reagents / Chemicals	Quantity
1	Dried yeast, labelled **yeast**	0.5 g
2	Sucrose powder, labelled **glucose**	1 g
3	250cm^3 beaker	1
4	Boiling tube	2
5	Bung and delivery tube	2
6	Test tube	2
7	Test-tube rack	1
8	10 cm^3 syringe	1
9	Styrofoam cup	1
10	Glass rod	1
11	Thermometer	1
12	Stop watch	1
13	Marker pen	1
14	Hot water at 60 ºC	2 tanks per lab
15	Toothpick	2
16	Clean slide	2
17	Cover slip	2
18	Sun shade slide	1
19	Stage micrometer 0.01mm	1
20	Pipe cleaner	2 each color
21	White tie	1
22	Label	4